Chapter 18, Problem 63P

Problem

For a given locality, calculate the number of stations allowable in the AM broadcasting band (540–1600 kHz) without interference with one another.

Step-by-step solution

Step 1 of 1

Space available for the AM channel, 540 KHz to 1600 KHz,

i.e.

Total space = 1600 KHz – 540 KHz = 1060 KHz

Now, the number of channels in 1060 KHz space, if all channels spaced 10 KHz apart

$$= \frac{1060 KH_z}{10 KH_z} \qquad \begin{bmatrix} \text{Without interference, carrier for} \\ \text{AM radio stations are spaced} \\ \text{10 KH}_z \text{ apart} \end{bmatrix}$$

$$= 106$$

$$or, \qquad \boxed{106 \text{ Stations}}$$